

WEIKAI LI

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EDUCATION

UCLA, Los Angeles, California (Ph.D. in Computer Science) 09/2023 – present

- GPA: 4.00/4.00
- Advisor: I am fortunate to be advised by [Prof. Yizhou Sun](#). I also closely work with [Prof. Jason Cong](#).
- Research areas: knowledge-augmented AI for chip design, Graph Neural Networks (GNNs), Large Language Models (LLMs).

Tsinghua University, Beijing, China (B.Eng in Computer Science) 08/2019 – 06/2023

- GPA: 3.92/4.00 (ranking: 10/195)
- Outstanding Undergraduate Graduate of Tsinghua University (top 10%)
- Advisors: I was fortunate to be advised by [Prof. Jie Tang](#), [Prof. Yuxiao Dong](#), and [Prof. Juanzi Li](#).
- Research areas: Graph Neural Networks (GNNs), Large language models (LLMs).

RESEARCH EXPERIENCES

Applied Scientist Intern at Amazon AWS (Graph ML Team) 06/2025 – present

Manager: Xiang Song; Mentor: Han Xie

- **LLM for GNN explanation:** I am developing LLM-based frameworks to generate natural language explanations for GNN predictions.

Graduate Student Researcher at UCLA 09/2023 – present

Advised by Prof. Yizhou Sun; also closely working with Prof. Jason Cong

- **Knowledge-augmented AI for HLS prediction:** I enhanced domain generalizability of GNNs for high-level synthesis (HLS) prediction with a hierarchical mixture-of-experts (MoE) structure based on domain knowledge. Besides, I design a hybrid model integrating hardware analytical model with data-driven AI model.
- **Understanding LLMs' internal mechanism:** We analyzed the influence of LLM's post-training on its internal mechanisms across four perspectives.

Undergraduate Student Researcher at Tsinghua University 01/2023 – 07/2023

Advised by Prof. Juanzi Li

- **Analyzing LLMs' limitations:** I analyzed the limitations of LLM's in-context learning on the event relation extraction task and demonstrated the importance of finetuning.

Undergraduate Student Research Intern at UCLA 07/2022 – 12/2022

Advised by Prof. Yizhou Sun

- **Node Influence Estimation on GNN predictions:** I designed an efficient approximation method to estimate the node influence on a GNN model's prediction.

Undergraduate Student Researcher at Tsinghua University 08/2021 – 07/2022

Advised by Prof. Jie Tang and Prof. Yuxiao Dong

- **GNN benchmark:** We discovered the phenomenon of over-tuning hyper-parameters in GNN model evaluation and proposed a new benchmark to improve stable evaluation.

PUBLICATIONS

- [1] Hongzhe Du*, **Weikai Li***, Min Cai, Karim Saraipour, Zimin Zhang, Himabindu Lakkaraju, Yizhou Sun, Shichang Zhang. “How Post-Training Reshapes LLMs: A Mechanistic View on Knowledge, Truthfulness, Refusal, and Confidence.” COLM 2025 & Outstanding Paper Award at the New England NLP Workshop 2025.
- [2] Zijian Ding, Tung Nguyen, **Weikai Li**, Aditya Grover, Yizhou Sun, Jason Cong. “Iceberg: Enhancing HLS Modeling with Synthetic Data.” LAD 2025.
- [3] **Weikai Li**, Ding Wang, Zijian Ding, Atefeh Sohrabizadeh, Zongyue Qin, Jason Cong, and Yizhou Sun. “Hierarchical Mixture of Experts: Generalizable Learning for High-Level Synthesis.” AAAI 2025.
- [4] Zijian Ding, Atefeh Sohrabizadeh, **Weikai Li**, Zongyue Qin, Yizhou Sun, and Jason Cong. “Efficient Task Transfer for HLS DSE.” ICCAD 2024.
- [5] Yunsheng Bai, Atefeh Sohrabizadeh, Zijian Ding, Rongjian Liang, **Weikai Li**, Ding Wang, Haoxing Ren, Yizhou Sun, and Jason Cong. “Learning to Compare Hardware Designs for High-Level Synthesis.” MLCAD 2024.
- [6] **Weikai Li**, Zhiping Xiao, Xiao Luo, and Yizhou Sun. “Fast Inference of Removal-Based Node Influence.” WWW 2024.
- [7] Jifan Yu*, Xiaozhi Wang*, Shangqing Tu, Shulin Cao, Daniel Zhang-Li, Xin Lv, Hao Peng, Zijun Yao, Xiaohan Zhang, Hanming Li, Chunyang Li, Zheyuan Zhang, Yushi Bai, Yantao Liu, Amy Xin, Nianyi Lin, Kaifeng Yun, Linlu Gong, Jianhui Chen, Zhili Wu, Yunjia Qi, **Weikai Li**, Yong Guan, Kaisheng Zeng, Ji Qi, Hailong Jin, Jinxin Liu, Yu Gu, Yuan Yao, Ning Ding, Lei Hou, Zhiyuan Liu, Bin Xu, Jie Tang, and Juanzi Li. “KoLA: Carefully Benchmarking World Knowledge of Large Language Models.” ICLR 2024.
- [8] Hao Peng*, Xiaozhi Wang*, Jianhui Chen*, **Weikai Li**, Yunjia Qi, Zimu Wang, Zhili Wu, Kaisheng Zeng, Bin Xu, Lei Hou, and Juanzi Li. “When does In-context Learning Fall Short and Why? A Study on Specification-Heavy Tasks.” ArXiv preprint 2023.
- [9] Ziang Li*, Ming Ding*, **Weikai Li**, Zihan Wang, Ziyu Zeng, Yukuo Cen, and Jie Tang. “Rethinking the Setting of Semi-supervised Learning on Graphs.” IJCAI 2022.

(* equal contribution)

SELECTED AWARDS AND HONORS

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| • Amazon Trainium Fellowship (Fall 2025) | 2025 |
| • Outstanding Undergraduate Graduate of Tsinghua University (top 10%) | 2023 |
| • Comprehensive Excellence Scholarship for junior year (top 10%) | 2022 |
| • Comprehensive Excellence Scholarship for sophomore year (top 10%) | 2021 |
| • Academic Excellence Scholarship for freshman year | 2020 |
| • Runner-up in Tsinghua Innovation Competition, industry and technology track, freshman group | 2019 |
| • First Prize in the National High School Mathematics League | 2018 |

ACADEMIC SERVICES

- Conference reviewer: ICLR 2024, ICML 2024, ACMMM 2024, NeurIPS 2024, LoG 2024, AAAI 2025, ICLR 2025, WWW 2025, ICML 2025, ICCV 2025, NeurIPS 2025, COLM 2025 Workshop

- Journal reviewer: IEEE TBD 2024, PLOS ONE 2024
- Conference volunteer: NSF AI for EDA Workshop 2024

TEACHING EXPERIENCE

- 2024 fall, 2025 winter, and 2025 spring: TA of CS 31 (Introduction to Computer Science I), UCLA
- 2024 summer: TA of CS 97 (Introduction to Data Science), UCLA summer school for high school students